EXPERIMENT 10
To Implement LDA

Dataset Used

Iris Dataset — 150 samples, 4 numerical features and 3 classes: setosa, versicolor and virginica.

Program

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# ============================================================
# EXPERIMENT 10
# TO IMPLEMENT LDA
# Dataset: Iris Dataset
# ============================================================

import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

# 1. LOAD DATASET
iris = load_iris()

X = iris.data
y = iris.target

print("========== DATASET ==========")
print("Dataset Shape:", X.shape)
print("Features:", iris.feature_names)
print("Classes:", iris.target_names)

# 2. STANDARDIZE DATA
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("\\nStandardized Data Shape:", X_scaled.shape)

# 3. APPLY LDA
lda = LinearDiscriminantAnalysis(n_components=2)

X_lda = lda.fit_transform(X_scaled, y)

# 4. DISPLAY RESULTS
print("\\n========== LDA ==========")

print("Explained Variance Ratio:")
print(np.round(lda.explained_variance_ratio_, 4))

print("Total Variance Retained:")
print(round(lda.explained_variance_ratio_.sum(), 4))

print("Reduced Shape:")
print(X_lda.shape)

print("\\nFirst 5 LDA Values:")
print(np.round(X_lda[:5], 4))

# 5. VISUALIZATION
plt.figure(figsize=(7, 5))

plt.scatter(
    X_lda[:, 0],
    X_lda[:, 1],
    c=y
)

plt.xlabel("Linear Discriminant 1")
plt.ylabel("Linear Discriminant 2")
plt.title("LDA - Iris Dataset")

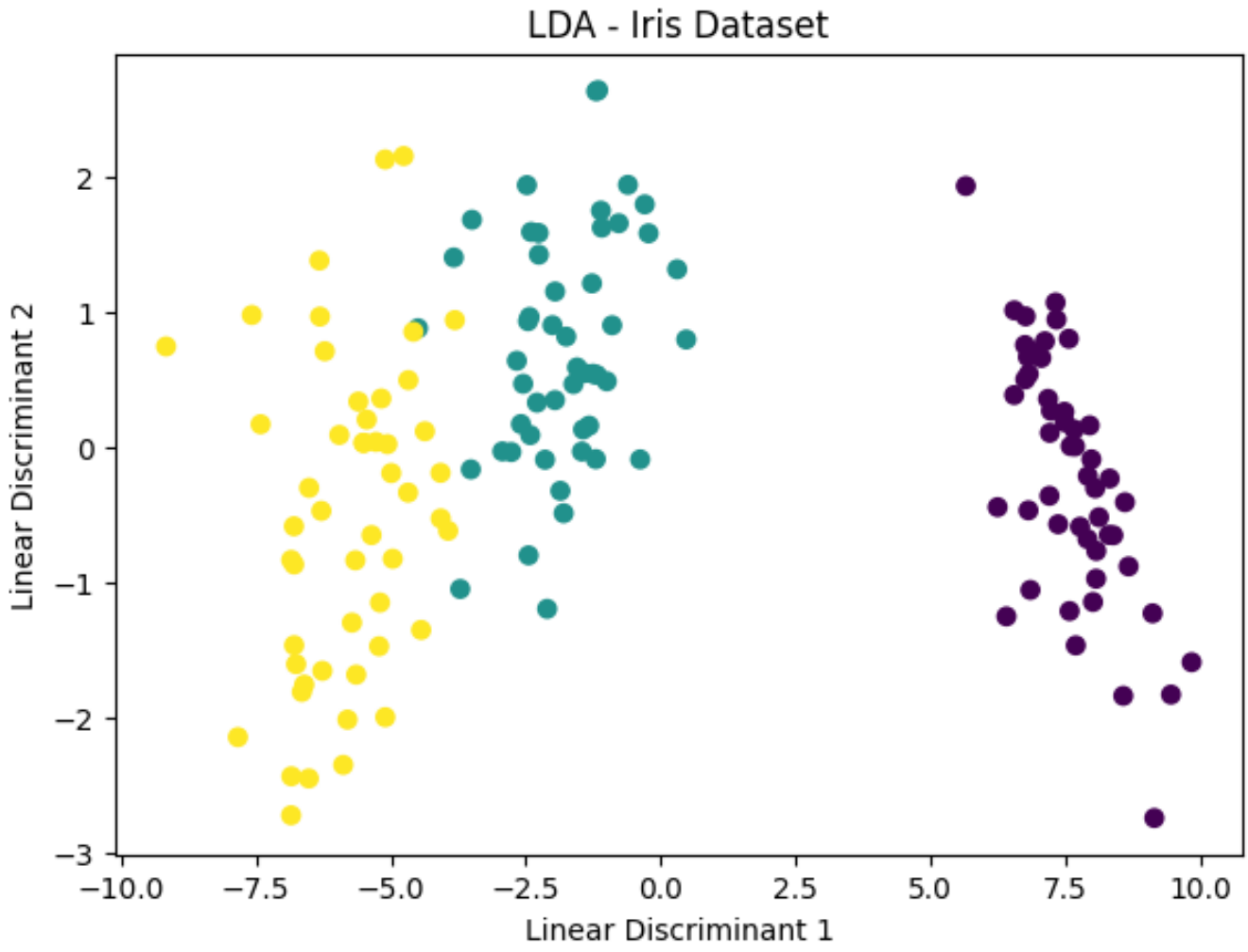
plt.show()
```

Output

```
========== DATASET ==========
Dataset Shape: (150, 4)
Features: ['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
Classes: ['setosa' 'versicolor' 'virginica']

Standardized Data Shape: (150, 4)

========== LDA ==========
Explained Variance Ratio:
[0.9912 0.0088]
Total Variance Retained:
1.0
Reduced Shape:
(150, 2)

First 5 LDA Values:
[[ 8.0618 -0.3004]
 [ 7.1287  0.7867]
 [ 7.4898  0.2654]
 [ 6.8132  0.6706]
 [ 8.1323 -0.5145]]
```

LDA Graph